
Deep Deterministic Policy Gradient (DDPG)

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CMPM118 Spring 2026

What is DDPG?

1

Definition

DDPG is a reinforcement learning model built off of the principles of a Deep Q Network to solve the problem of using deep RL to efficiently simulate continuous action simulations.

2

Related to DQN

DQN, or Deep Q Network, is a deep neural network that uses Q-Learning and a memory buffer for efficient reinforcement learning. DQN uses the principles of a memory buffer to continuously learn from previous iterations, applied towards discrete action simulations like Atari games.

Why DDPG?

1

Continuous

DDPG solves the issue of deep reinforcement learning for continuous actions, rather than just discrete actions like its predecessor DQN. For actions like how much of the wheel to turn, a DDPG is better suited.

2

Memory Buffer

DDPG adopts its predecessor's principle of a memory buffer, which randomly samples previous iterations of the training timesteps and learns from them additionally.

3

Stable Target Values

DDPG creates copies of its actor and critic networks in order to create better, more stable target values that it can direct itself towards. This is better than previous models where target values can dramatically shift based on the state of the model.

The Algorithm

1

Initialize

Randomly initialize the actor & critic networks, target network, memory buffer

2

Episodes

For each episode:

- Observe the environment

3

Timestep

For each timestep within an episode:

Execute a selected action based on the policy

Store the reaction into the memory buffer and relearn from random buffer samples

4

Update

For each timestep within an episode (cont).

- Update the actor & critic networks

Results

1

Simulations

Tested on numerous environments:
blockworld, canada, cartpole, cheetah, fixedReacher, gripper, hopper, hyq, pendulum, reacher, walker2d

2

Atari

- More efficient at Atari games than DQN
- Solved all tested Atari games within 20 million steps

3

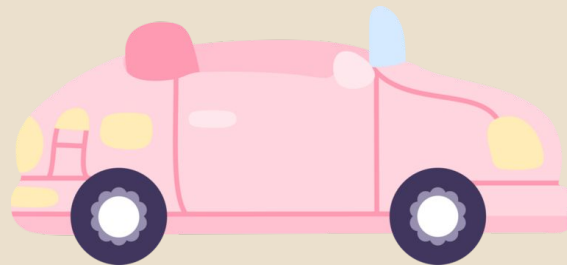
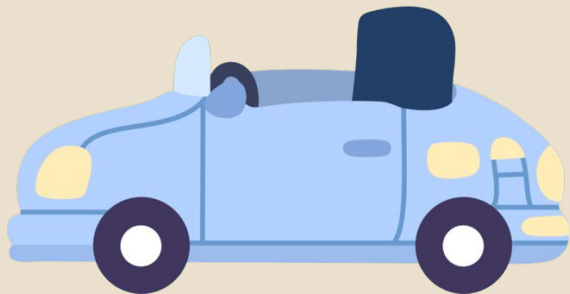
Torcs

Torcs is a racing game with three main driving actions: acceleration, braking, and steering

- Able to drive a full circuit a majority of the simulations

Applying to AVs

- Elements of car driving simulations are continuous rather than discrete, such as speed, braking, and other car functions.
- Having a model that can successfully model continuous actions efficiently is crucial to successful simulations that resemble real-life driving experiences
- A memory buffer is similar to how a driver learns from when they drive; they draw on their past experiences as well as fitting the journey they're on currently
- The paper is also tested on their custom racing game, which is similar to racing simulations that we conduct.



Thank You !

[1] T. P. Lillicrap et al.,
“Continuous control with deep
reinforcement learning,” arXiv.org,
<https://arxiv.org/abs/1509.02971>